Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

THIRD SEMESTER B.TECH. (CIVIL ENGINEERING)

END SEMESTER EXAMINATIONS, NOVEMBER 2024

ENGINEERING MATHEMATICS-III [MAT-2125]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 29 NOVEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

Q.NO	Questions	Marks	CO	BTL
1A.	Represent the function $f(x) = x^2,\ -\pi \le x \le \pi$, $f(x+2\pi) = f(x)$ as Fourier series.	4	1	3
1B.	Given $P(A) = \frac{1}{4}, P(B) = \frac{1}{3}$ and $P(A \cup B) = \frac{1}{2}$. Evaluate $P\left(A/\bar{B}\right)$ and $P(A \cap \bar{B})$.	3	2	4
1C.	A two-dimensional random variable (X, Y) has the joint probability distribution function $f(x,y) = kxy, 0 < x < y < 1$ find 'k' and check whether X and Y are independent.	3	3	3
2A.	Check whether the given vector $\vec{F} = (x+2y+4z)i + (2x-3y-z)j + (4x-y+2z)k$ is irrotational and find its scalar potential.	4	4	4
2B.	Obtain half-range cosine series expansion of $f(x) = x(l-x), 0 \le x \le l$.	3	1	4
2C.	Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$.	3	4	3
3A.	An office has 4 secretaries handling 20%, 60%,15% and 5% respectively of the files of certain reports. The probability that they misfile such reports is respectively 0.05, 0.1, 0.1, and 0.05. find the probability that the misfiled report is caused by the first secretary.	4	2	3
3B.	Solve $u_{xy} - u_{yy} = 0$ using the transformation $v = x, z = x + y$.	3	5	4

3C.	Show that $\nabla r^n = nr^{n-2}\vec{r}$, where r is a magnitude of position vector and n is a constant.	**3**	**4**	**4**
4A.	A 2-dimensional random variable (X, Y) has the joint pdf $f(x,y) = \begin{cases} 2-(x+y), & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & Otherwise \end{cases}$ compute the correlation coefficient ρ_{XY}.	**4**	**3**	**4**
4B.	Evaluate $\iint_S \vec{F}.\hat{n}\, ds$ where $\vec{F} = 2xyi + yz^2j + xzk$ and S is the surface of the parallelepiped bounded by the planes $x = 0, x = 2, y = 0, y = 1, z = 0$ and $z = 3$.	**3**	**4**	**3**
4C.	Find the Fourier transform of $f(x) = e^{-a\|x\|}, a > 0$.	**3**	**1**	**4**
5A.	The probability distribution of a finite random variable X is given by the following table x_i: -2, -1, 0, 1, 2, 3 $p(x_i)$: 0.1, k, 0.2, $2k$, 0.3, k (a) Find k (b) Find $P\{-1 < X \le 2\}$ (c) Find standard deviation σ	**4**	**3**	**4**
5B.	Solve $4u_x + u_y = 3u$. Given $u(0,y) = 2e^{5y}$ using variable separable method.	**3**	**5**	**3**
5C.	Compute the constant term and first harmonic of the Fourier series of $f(x)$ given in the following table. $x°$: **0, 60, 120, 180, 240, 300** $f(x)$: **4, 8, 15, 7, 6, 2**	**3**	**1**	**3**